

Formula sheet for Quiz 2

Note to students: you will be handed a copy with the quiz, so nothing here needs memorizing. It lists more formulas than any one question needs, so deciding which applies is still your job.

A layer, and its shapes

one neuron	$z = \mathbf{w}^\top \mathbf{x} + b, \quad a = \sigma(z)$
a layer, one example	$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}, \quad \mathbf{a} = \sigma(\mathbf{z})$
a layer, a batch of N	$\mathbf{Z} = \mathbf{X}\mathbf{W}^\top + \mathbf{b}, \quad \mathbf{A} = \sigma(\mathbf{Z})$
$\mathbf{W} : (n_{\text{out}} \times n_{\text{in}})$, one row per unit	$\mathbf{b} : (n_{\text{out}})$, one entry per unit
$\mathbf{X} : (N \times n_{\text{in}})$, one row per example	$\mathbf{A} : (N \times n_{\text{out}})$
parameters in one layer	$n_{\text{out}} \times n_{\text{in}} + n_{\text{out}}$

The forward pass

$$\mathbf{a}^{(0)} = \mathbf{x}, \quad \mathbf{a}^{(\ell)} = \sigma^{(\ell)}(\mathbf{W}^{(\ell)}\mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)}), \quad \hat{\mathbf{y}} = \mathbf{a}^{(L)}$$

The superscript is a **layer index**, not a power.

Activations

$$\begin{aligned} \text{sigmoid } \sigma(z) &= \frac{1}{1 + e^{-z}} \in (0, 1) & \tanh(z) &= \frac{e^z - e^{-z}}{e^z + e^{-z}} \in (-1, 1) \\ \text{ReLU}(z) &= \max(0, z) & \text{softmax}(\mathbf{z})_k &= \frac{e^{z_k}}{\sum_j e^{z_j}}, \quad \sum_k = 1 \end{aligned}$$

All are applied **elementwise**, except softmax, which is applied across a whole layer.

Choosing the output layer

Problem	Output activation	Loss
Regression, one or several numbers	none (identity)	MSE or MAE
Binary classification	sigmoid	binary cross-entropy
Multi-class, one label	softmax	categorical cross-entropy

Losses

$$\begin{aligned} \text{MSE} &= \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2 & \text{MAE} &= \frac{1}{n} \sum_i |y_i - \hat{y}_i| \\ \text{BCE} &= -\frac{1}{n} \sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)] \end{aligned}$$

Gradients and training

$$\mathbf{w}_{\text{new}} = \mathbf{w}_{\text{old}} - \lambda \nabla \mathcal{L}(\mathbf{w}_{\text{old}}) \quad \text{iterations per epoch} = N_{\text{train}} / \text{batch size}$$

$$\partial \mathcal{L} / \partial w_j \approx [\mathcal{L}(\mathbf{w} + h \mathbf{e}_j) - \mathcal{L}(\mathbf{w})] / h \quad \text{costs } P + 1 \text{ forward passes for } P \text{ parameters}$$

epoch one full pass over the training set · **iteration** one update, from one batch · **batch size** examples per update · **learning rate** λ step size